

Chapter 16.

§16.1.

Vector fields

2D: $F(x, y) = \langle f_1(x, y), f_2(x, y) \rangle$ — two variables, output is a 2D vector

3D: $F(x, y, z) = \langle f_1(x, y, z), f_2(x, y, z), f_3(x, y, z) \rangle$
— three variables, output is a 3D vector.

Gravitational field:

$$F(\vec{X}) = \frac{mMG}{r^2} \left(-\frac{\vec{X}}{|\vec{X}|} \right) = -\frac{mMG}{|\vec{X}|^3} \vec{X} \quad \text{where } \vec{X} = (x, y, z).$$

Gradient field.

∇f , where f is a differentiable scalar function in $\mathbb{R}^d$.

$$d=2: \nabla f = f_x(x, y)\mathbf{i} + f_y(x, y)\mathbf{j}$$

$$d=3: \nabla f = f_x(x, y, z)\mathbf{i} + f_y(x, y, z)\mathbf{j} + f_z(x, y, z)\mathbf{k}$$

Conservative Vector field

$$F = \nabla f$$

↳ potential function of the vector field F

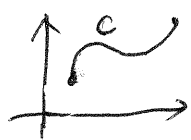
↳ gradient field of f , is conservative ($\oint_C F \cdot d\mathbf{r} = 0$).

for example: The gravitational field is conservative since its potential

$$\text{function is } f(x, y, z) = \frac{mMG}{\sqrt{x^2 + y^2 + z^2}}$$

§16.2.

Curve $C: x = x(t), y = y(t), a \leq t \leq b$



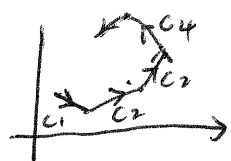
The line integral of the function f along C is

$$\int_C f(x, y) ds = \int_a^b f(x(t), y(t)) \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

If $f(x, y) = 1$, then it is the length of the curve

$$L = \int_C ds = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

Piece-wise smooth function $\{C_i\}_{i=1}^n = C$



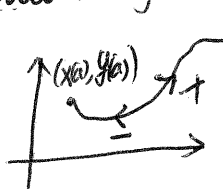
$$\int_C f(x,y) ds = \int_{C_1} f(x,y) ds + \dots + \int_{C_n} f(x,y) ds$$

The line integrals with respect to x and y .

$$\begin{cases} \int_C f(x,y) dx = \int_a^b f(x(t), y(t)) x'(t) dt \\ \int_C f(x,y) dy = \int_a^b f(x(t), y(t)) y'(t) dt \end{cases}$$

(Remark: they are dependent to the direction of C)

orientation of a curve: $x=x(t), y=y(t), a \leq t \leq b$



positive direction: increasing values of the parameter t .

$$\begin{cases} \int_{-C} f(x,y) dx = - \int_C f(x,y) dx \\ \int_{-C} f(x,y) dy = - \int_C f(x,y) dy \end{cases}$$

$$\text{but } \int_{-C} f(x,y) ds = \int_C f(x,y) ds$$

$\Delta x, \Delta y$ has signs

ds is always positive.

Line integral in space

Curve $C: x=x(t), y=y(t), z=z(t), a \leq t \leq b$

$$\int_C f(x,y,z) ds = \int_a^b f(x(t), y(t), z(t)) \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} dt = \int_a^b f(t) |r'(t)| dt$$

Line integrals of vector fields

vector field: $F(x,y,z)$

curve $C: r(t), a \leq t \leq b$

$$\boxed{\int_C F \cdot dr = \int_a^b F(r(t)) \cdot r'(t) dt} = \int_C F \cdot \underbrace{T}_{\substack{\text{tangential vector} \\ \frac{r'(t)}{|r'(t)|}}} ds = \int_a^b F \cdot \frac{r'(t)}{|r'(t)|} |r'(t)| dt$$